

Modular Operator Discovery TSP Report

Overview

- Condition count: 7.
- Best final transfer gap: phase7_modular_operator_evolution.
- Best held-out TSPLIB gap: phase7_full_solver_evolution.
- Surviving modular candidates: none.

Run Metadata

- run_name: run_20260520_042327_l
- started_at_local: 2026-05-20 04:23:27
- finished_at_local: 2026-05-20 04:57:42
- duration_hhmm: 00:34
- duration_seconds: 2055.357
- seed_offset: 11000
- replicate_label: l
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_baseline_heuristic_only	baseline_only	none	score_only	0.235058	0.054968	0.121317	0.0	0.0	False	0.0	0.0
phase7_full_solver_evolution	full_solver	none	score_only	0.213387	0.064916	0.159398	0.819776	0.56	False	0.0	0.0
phase7_modular_operator_evolution	modular_operator	none	score_only	0.235058	0.046321	0.115856	0.895543	0.44	False	0.008978	0.669401
phase7_modular_operator_random_replay	modular_operator	random	score_only	0.235058	0.054968	0.121317	0.888046	0.44	False	-0.000133	0.013898
phase7_modular_operator_diversity_residual_replay	modular_operator	diversity_residual	score_only	0.236585	0.064041	0.12761	0.899613	0.445	False	0.002398	0.006216

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_modular_operator_compression_pressure	modular_operator	none	novelty_gate	0.235058	0.050827	0.118702	0.892386	0.44	False	0.008043	0.684063
phase7_modular_operator_pareto_selection	modular_operator	none	pareto	0.235667	0.054968	0.121542	0.887354	0.4375	False	0.009335	0.355837

Condition Notes

phase7_baseline_heuristic_only

- Execution mode baseline_only on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.0, complexity 0.0, and adaptation efficiency 0.0.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_full_solver_evolution

- Execution mode full_solver on host scaffold whole_solver.
- Final gaps: TSPLIB 0.213387, family holdout 0.064916, combined 0.159398.
- Accepted novelty 0.819776, complexity 0.56, and adaptation efficiency -0.03038.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_modular_operator_evolution

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.046321, combined 0.115856.
- Accepted novelty 0.895543, complexity 0.44, and adaptation efficiency 0.009694.
- Last-epoch runtime 57.114525 ms and distance evaluations 4782.75.
- Validation: surviving False, transplant delta 0.008978, positive scaffolds 1, Pareto runtime inflation 0.669401.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.767933, "gap_delta": -0.002615, "runtime_inflation": 0.669401, "same_gap_faster": false}

phase7_modular_operator_random_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.888046, complexity 0.44, and adaptation efficiency 0.006187.
- Last-epoch runtime 67.78975 ms and distance evaluations 3810.75.

- Validation: surviving False, transplant delta -0.000133, positive scaffolds 1, Pareto runtime inflation 0.013898.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 11, "complexity_score": 0.42, "distance_eval_inflation": 0.0, "gap_delta": 0.0, "runtime_inflation": 0.013898, "same_gap_faster": false}

phase7_modular_operator_diversity_residual_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.236585, family holdout 0.064041, combined 0.12761.
- Accepted novelty 0.899613, complexity 0.445, and adaptation efficiency 0.003117.
- Last-epoch runtime 96.743475 ms and distance evaluations 3810.75.
- Validation: surviving False, transplant delta 0.002398, positive scaffolds 0, Pareto runtime inflation 0.006216.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 13, "complexity_score": 0.46, "distance_eval_inflation": 0.003718, "gap_delta": 0.006293, "runtime_inflation": 0.006216, "same_gap_faster": false}

phase7_modular_operator_compression_pressure

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.050827, combined 0.118702.
- Accepted novelty 0.892386, complexity 0.44, and adaptation efficiency 0.016267.
- Last-epoch runtime 245.747425 ms and distance evaluations 7156.75.
- Validation: surviving False, transplant delta 0.008043, positive scaffolds 1, Pareto runtime inflation 0.684063.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.803776, "gap_delta": -0.003304, "runtime_inflation": 0.684063, "same_gap_faster": false}

phase7_modular_operator_pareto_selection

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235667, family holdout 0.054968, combined 0.121542.
- Accepted novelty 0.887354, complexity 0.4375, and adaptation efficiency 0.013158.
- Last-epoch runtime 75.819425 ms and distance evaluations 3810.75.
- Validation: surviving False, transplant delta 0.009335, positive scaffolds 1, Pareto runtime inflation 0.355837.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.278237, "gap_delta": 0.000224, "runtime_inflation": 0.355837, "same_gap_faster": false}

Judge Appendix

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TSP Modular-Operator Discovery Suite Review

Summary of Key Metrics and Focus

Condition	Held-out TSP LIB Gap	Family Holdout Gap	Transplant Mean Gap Delta	Ablation Validation	Surviving Candidate	Operator Type	Comments on Transfer / Reuse
phase7_baseline_heuristic_only	0.235058	0.054968	0.0	N/A	No	N/A	Baseline non-modular heuristic.
phase7_full_solver_evolution	**0.213387**	0.064916	0.0	N/A	No	N/A	Best TSPLIB gap overall but no modular operator identified or validated.
phase7_modular_operator_evolution	0.235058	**0.046321**	0.008978	Yes (~0.0026 gap diff)	No	Restart Controller	Small family gap improvement. Transplant failed on key scaffolds (sparse_three_opt, clustered_local_search). Ablation shows minimal effect but confirms operator effect. Runtime inflation high (0.67x).
phase7_modular_operator_random_replay	0.235058	0.054968	-0.000133	No (ablation shows no effect)	No	Acceptance	No family improvement or transfer benefit. Ablation disables operator with zero effect. Non-robust operator.
phase7_modular_operator_diversity_residual_replay	0.236585	0.064041	0.002398	No (ablation disables operator improves gap)	No	Candidate Ranker	Family gap worsens significantly on grid_like_tsp. No positive family signals. Transplant failed in multiple scaffolds. Ablation does not confirm operator efficacy.
phase7_modular_operator_compression_pressure	0.235058	0.050827	0.008043	Yes (~0.0033 gap diff)	No	Restart Controller	Similar operator to modular_operator_evolution condition. Provides small family gap improvement on two_cluster_bottleneck_tsp. Transplant issues and high runtime inflation (~0.68x). Operator validated by ablation but not surviving transplant.
phase7_modular_operator_pareto_selection	0.235667	0.054968	0.009335	No (ablation no effect)	No	Candidate Pruner	No family-level improvement. Ablation indicates operator effect negligible or harmful. Runtime inflation moderately high (0.36x). Transplant mostly negative or neutral.

Conservative Interpretation & Prioritization

Transfer Evidence (Held-out TSPLIB and Family Gaps)

- No modular operators improve held-out TSPLIB gap beyond baseline (best TSPLIB gap of 0.213387 comes from full solver evolution but no modular operator identified).
- Family holdout gap gains are marginal (~ 0.046 – 0.050 vs baseline ~ 0.054), only seen in two restart-controller operators (modular_operator_evolution, compression_pressure).

Transplant & Ablation Validation

- Only restart-controller operators (phase7_modular_operator_evolution and compression_pressure) show positive ablation signals validating operator effect.
- Ablation gap delta vs full operator ~ 0.0026 to 0.0033 (small but consistent).
- Neither restart-controller operator survives clean transplant tests cleanly:
- Both fail in sparse_three_opt and clustered_local_search.
- Transplant gap deltas positive (performance worsens or no improvement).
- High runtime inflation (~ 0.67 – 0.68) severely limits practical reusability.
- Other operators (acceptance, candidate ranker, candidate pruner):
- Fail ablation tests (disabling operator causes no harm or slight improvement).
- Show no family-level or TSPLIB improvement.
- Fail transplant validation or show neutral-to-negative transfer effects.
- No surviving candidates.

Pareto Tradeoffs

- Restart-controller operators cause notable runtime inflation ($\sim 66\%$ – 68%), suggesting poor efficiency tradeoff.
- Candidate pruner operator has moderate runtime inflation ($\sim 36\%$) but no robust benefit and failed ablation.
- No operator achieves better gap at same or faster runtime.

Operator Reusability and Structure Claims

- Restart-controller operators embody a deterministic restart mechanism tuned for EUC2D TSP stagnation scenarios, enabling escapes from traps.
- However, these operators are insufficiently robust in diverse scaffolds and fail transplant conditions.
- Other operators do not convincingly surpass baseline or validate as modular reusable components.
- No operators meet surviving candidate threshold.

Algorithm Discovery Claims

- No operator passes both the scaffold and family holdout tests to justify claiming algorithm discovery.
- Best modular candidates (restart controllers) show marginal benefits but poor transfer robustness and high runtime costs.

Final Assessment

Operator Name	Meets Key Criteria?	Comments
det_restart_controller_stagnation_balanced_euc2d (restart_controller)	**No** (fails transplant and family tests)	Validated by ablation; marginal family gains; fails transplant on multiple scaffolds; high runtime.
deterministic_restart_with_perturbation_escape (restart_controller)	**No** (same as above)	Same profile as above; slight family improvement; runtime inflation; no survival.
bounded_worse_threshold_acceptance_stagnation_bias (acceptance)	**No** (fails ablation and transfer)	No ablation effect; no family gain; no transplant improvement.
candidate_ranker_bottleneck_grid_aware_reorder (candidate_ranker)	**No** (negative family and no ablation)	Fails ablation; negative gap delta on grid_like family; no transplant success.
candidate_pruning_bottleneck_cluster_safe (candidate_pruner)	**No** (no ablation effect, no family gain)	Runtime inflation; no positive family or transplant signals.

Key Recommendations

- ****No** modular operator currently qualifies as a reusable, transportable component under transplant and family holdout tests.
- The restart controller operators show some promise for modular restart mechanisms tuned for EUC2D stagnation but require improved transplant robustness and runtime efficiency.
- Future work should focus on reducing runtime inflation and validating operators across more scaffolds for reusable designs.
- Full solver evolution outperforms all modular operators in TSPLIB benchmarks but does not isolate reusable operator structures.
- Avoid claims of algorithm discovery until surviving candidate conditions are met (scaffold and family holdout validated).

Summary

The suite's main transfer evidence lies in phase7_modular_operator_evolution (restart controller) and compression_pressure (restart controller) operators, both validated by ablation but failing robust transplant and family holdout tests, limiting claims to reusable modular operator structure. No modular operator improves held-out TSPLIB gap significantly nor survives transplant sufficiently to justify claiming algorithm discovery. The full-solver evolution condition performs best on TSPLIB but without modular operator discovery.